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1. Make the Dataset using the mobile phone

The screenshot displays the Edge Impulse Studio web interface. On the left is a sidebar with navigation options: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Impulse design. The main area has a top navigation bar with tabs for Dataset, Data explorer, Data sources, Synthetic data, AI labeling (marked as NEW), and CSV Wizard. Below this, a summary section shows 'DATA COLLECTED 30 items' and 'TRAIN / TEST SPLIT 80% / 20%'. The 'Dataset' section is active, showing a grid of training samples. The first sample is labeled 'Kishan.6egfvi8t' with a 'Monitor' label. The next two samples are labeled 'Bottle'. The bottom row shows samples labeled 'Boottle1.6ebbpnot', 'Boottle.6ebbpnki', and 'Notebook4.6ebbonbl'. On the right, there is a 'Collect data' section with a 'Connect a device' button and a 'RAW DATA' section with a 'Click on a sample' button. An 'Upgrade Plan' banner is visible at the bottom left of the main area.

EDGE IMPULSE

Dataset | Data explorer | Data sources | Synthetic data | AI labeling **NEW** | CSV Wizard

DATA COLLECTED
30 items

TRAIN / TEST SPLIT
80% / 20%

Collect data

Connect a device to start collecting data

RAW DATA
Click on a sample

Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Impulse design

Create Impulse
Image
Transfer learning

Upgrade Plan
Get access to higher job limits and more collaborators.
View plans

Dataset

Training (24) | Test (6)

Kishan.6egfvi8t
LABEL: Monitor

Bottle
LABEL: Bottle

Bottle
LABEL: Bottle

Boottle1.6ebbpnot

Boottle.6ebbpnki

Notebook4.6ebbonbl

<https://studio.edgeimpulse.com/studio/872543/acquisition/training?page=1#>

Dataset train / test split ratio

×

Training data is used to train your model, and **testing data** is used to test your model's accuracy after training. We recommend an approximate 80/20 train/test split ratio for your data for every class (or label) in your dataset, although especially large datasets may require less testing data.

SUGGESTED TRAIN / TEST SPLIT

80% / 20%

Labels in your dataset ?

BOTTLE

80% / 20% (4 / 1)

FAN

80% / 20% (4 / 1)

KEYBOARD

80% / 20% (4 / 1)

MONITOR

80% / 20% (4 / 1)

MOUSE

80% / 20% (4 / 1)

NOTEBOOK

80% / 20% (4 / 1)

Dismiss

2. Create Impulse for Transfer Learning

The screenshot displays the Edge Impulse IDE interface. On the left is a sidebar with navigation options: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, and Transfer learning. Below these is an 'Upgrade Plan' button. The main workspace contains four panels:

- Image data** (red background):
 - Input axes: image
 - Image width: 96
 - Image height: 96
 - Resize mode: Fit shortest
- Image** (white background):
 - Name: Image
 - Input axes (1): Image
- Transfer Learning (Images)** (purple background):
 - Name: Transfer learning
 - Input features: ☒ Image
 - Output features: 6 (Bottle, Fan, Keyboard, Monitor, Mouse, Notebook)
- Output features** (teal background):
 - 6 (Bottle, Fan, Keyboard, Monitor, Mouse, Notebook)

A 'Save Impulse' button is located at the bottom right of the Output features panel. A status bar at the top of the workspace reads: 'An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.'

3. Go to Image their General Feature

Training set

Data in training set

24 items

Classes

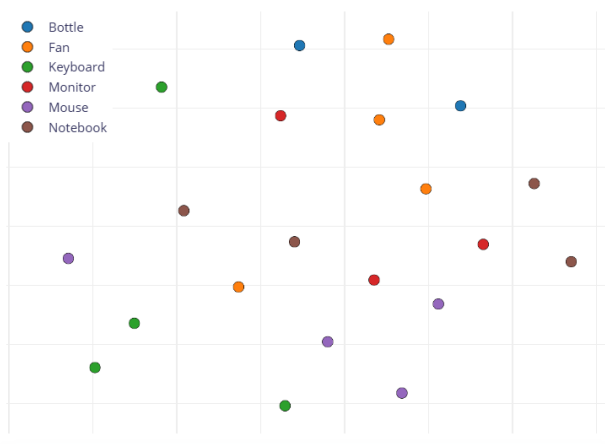
6 (Bottle, Fan, Keyboard, Monitor, Mouse, Notebook)

Generate features

Feature generation output

🔊 (0)

Feature explorer



On-device performance ?

PROCESSING TIME

1 ms.

PEAK RAM USAGE

4 KB

Resume tutorial

4. Go to Transfer Learning then save and train the model

Training settings

Number of training cycles ?

20

Use learned optimizer ?

☐

Learning rate ?

0.0005

Training processor ?

GPU

Data augmentation ?

☐

Advanced training settings

Neural network architecture

Input layer (27,648 features)

MobileNetV2 96x96 0.35 (final layer: 16 neurons, 0.1 dropout)

Choose a different model

Model

Model version: ? Quantized (int8)

Last training performance (validation set)

ACCURACY

40.0%

LOSS

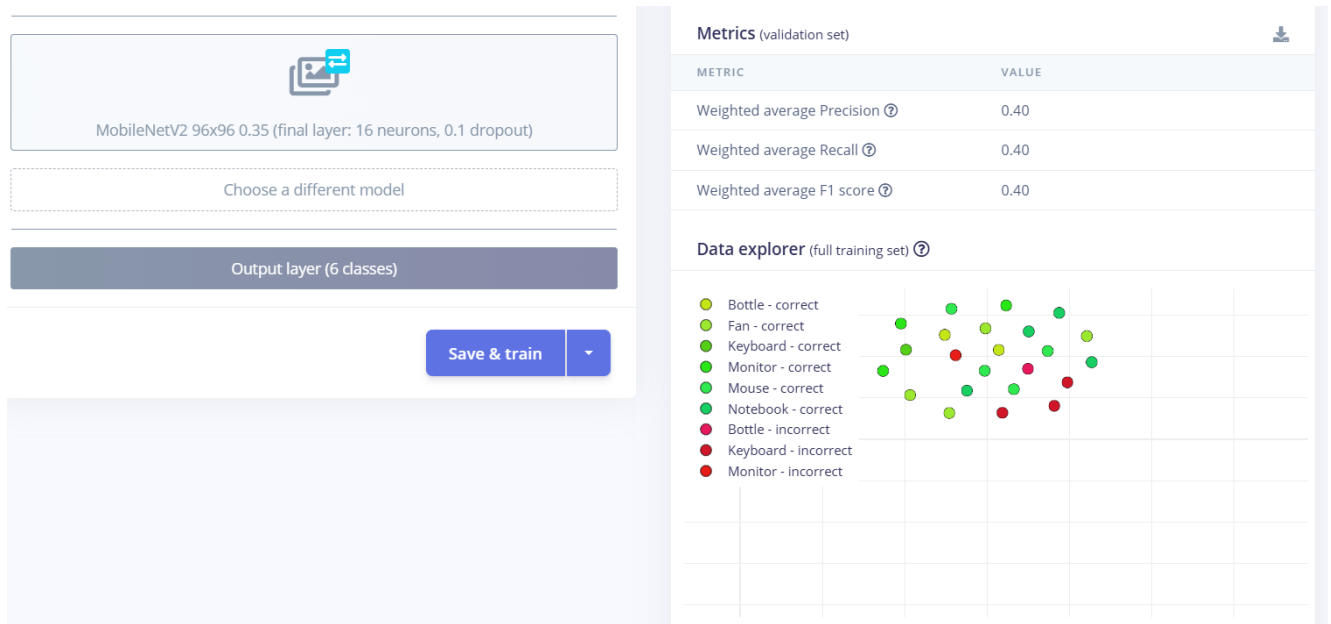
0.94

Confusion matrix (validation set)

	BOTTLE	FAN	KEYBOARD	MONITOR	MOUSE	NOTEBOOK
BOTTLE	-	-	-	-	-	-
FAN	0%	100%	0%	0%	0%	0%
KEYBOARD	33.3%	0%	0%	0%	0%	66.7%
MONITOR	-	-	-	-	-	-
MOUSE	0%	0%	0%	0%	100%	0%
NOTEBOOK	-	-	-	-	-	-
F1 SCORE	0.00	1.00	0.00		1.00	0.00

Metrics (validation set)

METRIC	VALUE
Weighted average Precision ?	0.40
Weighted average Recall ?	0.40
Weighted average F1 score ?	0.40



For Classification Learning

1. Create the Impulse for classification

The interface is divided into four main sections:

- Image data:** Contains input axes (image), image width (96), image height (96), and a resize mode dropdown set to "Fit shortest".
- Image:** Contains a name field (Image), input axes (1), and a dropdown menu (image).
- Classification:** Contains a name field (Classifier), input features (Image), and output features (6 (Bottle, Fan, Keyboard, Monitor, Mouse, Notebook)).
- Output features:** Contains a dropdown menu (6 (Bottle, Fan, Keyboard, Monitor, Mouse, Notebook)) and a "Save Impulse" button.

A "Resume tutorial" button is located at the bottom right.

2. Generate Image in the image section

Training set

Data in training set

24 items

Classes

6 (Bottle, Fan, Keyboard, Monitor, Mouse, Notebook)

Generate features

Feature generation output

(0)

Feature explorer

Bottle

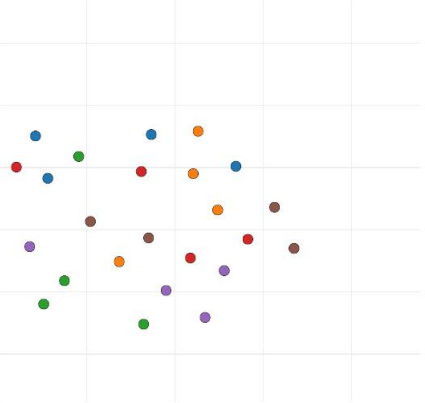
Fan

Keyboard

Monitor

Mouse

Notebook



On-device performance ⓘ

PROCESSING TIME

1 ms.

PEAK RAM USAGE

4 KB

3. Go to Classifier and save the train the model

Neural Network settings

Training settings

Number of training cycles ⓘ

10

Use learned optimizer ⓘ

☐

Learning rate ⓘ

0.0005

Training processor ⓘ

GPU

Advanced training settings

Neural network architecture

Neural network

Transfer learning

Input layer (27,648 features)

Training output

Model

Model version: ⓘ

Quantized (int8)

Last training performance (validation set)

ACCURACY

0.0%

LOSS

2.18

Confusion matrix (validation set)

	BOTTLE	FAN	KEYBOARD	MONITOR	MOUSE	NOTEBOOK
BOTTLE	-	-	-	-	-	-
FAN	0%	0%	0%	100%	0%	0%
KEYBOARD	0%	0%	0%	0%	100%	0%
MONITOR	-	-	-	-	-	-
MOUSE	0%	0%	0%	100%	0%	0%
NOTEBOOK	-	-	-	-	-	-
F1 SCORE		0.00	0.00	0.00	0.00	

